Computers in Human Behavior Reports 22 (2026) 101130

Contents lists available at ScienceDirect

Computers in Human Behavior Reports

journal homepage: www.sciencedirect.com/journal/computers-in-human-behavior-reports

Cognitive offloading in student–AI collaboration: A longitudinal analysis of prompting strategies☆

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ARTICLE INFO

Keywords:
Human–AI collaboration
Co-occurrence network analysis
Synthetic data
Cognitive offloading

ABSTRACT

Prompting an AI is not a neutral act: it reflects how students think, plan, and offload cognitive effort. As large language models (LLMs) become increasingly integrated into higher education, understanding how students engage with these tools is essential. This study examines cognitive offloading through students' interactions with LLMs when generating social network datasets. A total of 281 prompts from 122 submissions in four assignments were analyzed using qualitatively coded prompts and Co-Occurrence Network Analysis. High-quality submissions demonstrated cohesive prompting patterns, integrating contextual details, instructions, and polite language, leading to fewer disagreements and more effective task guidance. Low-quality submissions were characterized by disagreement and direct instructions, with limited contextualization and little constructive engagement. The findings show how cognitive effort is distributed in the student–AI collaboration and how prompting strategies develop in multiple assignments.

1. Introduction

Although research has long examined how humans interact through digital technologies, understanding how they engage cognitively and socially with artificial intelligence (AI) remains limited. The rapid expansion of AI has introduced new forms of partnership in which learners and systems share cognitive work. Hybrid intelligence has therefore been proposed as a model in which humans and AI complement each other's abilities in complex thinking and problem-solving (Molenaar et al., 2019). Yet, little is known about the psychological processes that shape these interactions. In educational settings, generative AI can function as an interactive partner that supports sense-making, exploration, and conversational learning (Michel-Villarreal et al., 2023). It offers a flexible space for meaning construction and knowledge transfer (Sharples, 2023) and can personalize feedback to guide learners through formative activities (Baidoo-Anu & Owusu Ansah, 2023). These interactions introduce not only new learning opportunities, but also new cognitive behaviors.

Large language models (LLMs) such as ChatGPT and Copilot can enhance learning by acting as conversational partners, co-designers, or Socratic opponents, supporting knowledge exploration, feedback, and personalized learning experiences (Baidoo-Anu & Owusu Ansah, 2023; Michel-Villarreal et al., 2023; Sharples, 2023). In higher education, AI can facilitate adaptive tutoring, formative feedback, and collaborative learning, functioning alongside teachers and peers to support goal-setting and co-regulation (J. Lodge et al., 2023). A central psychological question is how students distribute cognitive effort when working with AI. These advanced AI systems can act as external cognitive resources, often encouraging cognitive offloading rather than internal processing. When AI becomes an active collaborator in assignments, students may rely on it to generate ideas, structure solutions, or test reasoning. These forms of cognitive offloading can influence how students approach problem-solving and learning. Examining how students engage with AI collaboratively and how this reliance changes over time is essential for designing learning experiences that provide meaningful support while promoting active cognitive engagement.

To explore this dynamic, the present study analyses data from students who generated their own network datasets using LLMs across four assignments in a semester-long course. Qualitatively coded student prompts and Co-Occurrence Network Analysis (CNA) were employed to map the interactions between AI and learners and to examine how these evolving patterns differed between higher- and lower-quality submissions. In particular, the following research questions are addressed:

☆ This article is part of a Special issue entitled: 'Human-Centered Learning Analytics' published in Computers in Human Behavior Reports.

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https://doi.org/10.1016/j.chbr.2026.101130
Received 16 December 2025; Received in revised form 1 April 2026; Accepted 22 May 2026
Available online 27 May 2026
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1. Which prompt code combinations and centrality patterns characterize high- and low-quality submissions, and how do these reflect differences in cognitive offloading?
2. How do prompt code co-occurrence patterns develop across four successive assignments, and do high- and low-quality submissions diverge over time?

This paper is structured as follows. First, the literature on student–AI collaboration is reviewed, focusing on existing approaches to evaluating it, followed by an overview of prior research on Cognitive Load Theory in this context. Next, the dataset, data collection process, coding procedure, and analytical methods employed are described. Following that, the results of the co-occurrence network analysis are presented. Finally, the paper concludes with a discussion of the findings, their implications for educational practice, and directions for future research.

2. Background

This section is organized around the key concepts underpinning the study. Student–AI collaboration is examined first, followed by cognitive offloading and prompting as the mechanisms through which such collaboration is understood.

2.1. Student-AI collaboration

Research on generative AI in education spans questions of capability, what models can do, and questions of collaboration, how students work with AI to accomplish learning tasks. A significant strand of work continues to focus on benchmarking model performance (Zhang & Sun, 2025) and evaluating LLMs on specific educational tasks such as multiple-choice question generation (Awalurahman et al., 2025) or feedback provision (Yavariabdi et al., 2025). Alongside this, researchers have investigated the broader impact of generative AI on learning outcomes, with meta-analytic evidence establishing that AI-generated feedback is broadly comparable to human-provided feedback across performance and dispositional outcomes (Kaliisa et al., 2026), and that generative AI has consistently positive effects on cognitive, competency, and affective outcomes, with discipline type emerging as a meaningful moderator (Ma & Zhong, 2025). Building on this evidence base, attention has increasingly turned to a more granular question: not only whether AI can support learning, but also how students engage with it in practice.

Student–AI collaboration, understood here as the interactive, multi-turn exchange between a learner and an LLM system in the context of a structured academic task, has become a distinct object of inquiry. A first line of research has sought to identify recurring patterns in how students engage with AI. Studies point to the existence of a deep-engagement profile (Hao et al., 2026; Wang et al., 2023), a more efficient and results-oriented profile (Nguyen & Nguyen, 2025; Urban et al., 2025), and profiles varying in structure, exploration, or overall engagement level (Liu et al., 2024; Woollaston et al., 2025). These profiles, however, are largely static: they capture tendencies at an aggregate level and do not illuminate the moment-to-moment dynamics of collaboration.

A second line of research has addressed this by drawing on process-based methods such as epistemic network analysis, lag sequential analysis, transition network analysis, and sequence mining to compare the collaboration dynamics of student–AI interactions. These studies show that higher AI engagement and self-regulated learning are consistently linked to stronger outcomes (Lai et al., 2025; Urban et al., 2025), and that students who actively evaluate and extend AI output through critical thinking and higher-order cognition tend to outperform those who accept responses uncritically (Cong-Lem & Thuy-Dung, 2026; Wang et al., 2025; Zhao et al., 2026). Conversely, using AI primarily as an information source, delegating evaluative judgment to the AI, and engaging in lower-order cognitive processing are associated with weaker performance (López-Pernas et al., 2025a; Wang et al., 2025). The present study contributes to this line of work by examining the prompts students use when collaborating with an LLM across multiple assignments, with the aim of gaining insights into how cognitive effort is allocated and how prompting strategies relate to the quality of student output.

As generative AI becomes more deeply embedded in learning, there is increasing recognition that students' interactions with LLMs involve more than technical proficiency or procedural prompting skills. The way learners formulate prompts, interpret model responses, and decide when and how to revise their approach varies significantly across individuals and tasks (Lai et al., 2025; Urban et al., 2025). These patterns suggest that engagement with AI reflects deeper cognitive and metacognitive processes rather than simple tool use (Chen et al., 2025; López-Pernas et al., 2025a). In this regard, AI-generated feedback can influence how students monitor their work, make decisions, and adjust their strategies, meaning that understanding student–AI collaboration requires approaches that account for how learners navigate uncertainty, interpret AI output, and refine their strategies over time (An et al., 2025; Hao et al., 2026). While process-based studies have begun to map these dynamics, how regulatory and prompting behaviors are sustained, adapted, or transformed across a sequence of assignments remains an open question.

A particularly consequential source of uncertainty is the tendency of generative AI to produce outputs that are fluent and confident in presentation yet factually incorrect, logically inconsistent, or contextually inappropriate, a phenomenon widely described as *hallucinations* (Gu et al., 2024; Lane, 2025). Unlike errors in static reference materials, hallucinations lack obvious markers of uncertainty: models generate plausible text probabilistically based on learned patterns rather than verified knowledge, and confabulation tendencies lead them to fabricate details or citations to fill perceived information gaps (Elsayed, 2024). Empirical evidence confirms that even trained users struggle to reliably identify AI-generated errors (Zhou et al., 2025), and when students lack the foundational knowledge needed to recognize an incorrect response, hallucinations may increase cognitive load by generating confusion, requiring additional verification effort, or leading learners down unproductive problem-solving paths (Kim & Ji, 2026; Schulz et al., 2025). This is why evaluative judgment, the capacity to assess the quality and credibility of AI output against task-specific criteria defined a priori, is not merely a desirable disposition but a functional necessity for effective student–AI collaboration (Bearman et al., 2024; Cong-Lem & Thuy-Dung, 2026).

2.2. Cognitive load theory and student-AI collaboration

Cognitive Load Theory (CLT) is concerned with how human cognitive architecture processes learning materials and instructional complexity (Sweller, 1994, 2011a). CLT posits that learning is governed by the limited processing capacity of working memory, and that effective learning occurs only when cognitive demands do not exceed those limits (Sandoval-Medina et al., 2024; Sultanova, 2025). The total cognitive effort required during learning comprises three distinct types: intrinsic load, arising from the inherent complexity of the material; extraneous load, imposed by how information is presented; and germane load, reflecting the cognitive resources invested in constructing and refining schemas in long-term memory (Sultanova, 2025; Sweller, 1994). Effective learning requires managing intrinsic and extraneous loads so that sufficient working memory capacity remains available for germane processing (Kim & Ji, 2026; Sweller, 2011a).

The introduction of new technologies has long shaped how students offload cognitive effort, that is, the use of external resources to alter the information processing requirements of a task and reduce cognitive demand (Dawson, 2020). The Internet shifted learners' focus from memorizing content to locating it, which can reduce retention while

leaving learners equally or more confident in their understanding (Skulmowski, 2023). Graphing calculators enabled students to offload laborious computation, freeing attention for conceptual understanding and complex problem-solving (Roschelle & Singleton, 2008). Cognitive offloading can enhance learning by freeing resources for higher-order tasks, but its effects depend on task demands, perceived difficulty, and individual working memory capacity (Dawson, 2020; Skulmowski, 2023).

From a CLT perspective, generative AI represents a significant evolution in this offloading process. When used appropriately, AI can reduce extraneous load by simplifying tasks, thereby freeing resources for higher-order thinking (Cong-Lem & Thuy-Dung, 2026; Joo et al., 2026). However, excessive offloading risks reducing intrinsic load as well, by oversimplifying tasks and diminishing the need for germane processing (López-Pernas et al., 2025a; Wang & Luo, 2026). In such cases, students may complete tasks efficiently without engaging in the cognitive effort required for learning, producing what has been described as an illusion of mastery (Skulmowski, 2023; Wang & Luo, 2026). This risk is compounded by the interactive nature of generative AI: unlike passive tools, LLMs enable the generation of complete solutions instantly, which can reduce the motivation for critical evaluation and reflection (Bearman et al., 2024; Chen et al., 2025). Offloading can improve immediate performance and task efficiency, but may impair memory, problem-solving skills, and independent reasoning over time, posing a direct challenge to educational goals of deep engagement and long-term retention (Grinschgl et al., 2021; Schulz et al., 2025).

In the context of student–AI collaboration, cognitive offloading is operationalized through prompt content: prompts that delegate generative, evaluative, or structuring work to the AI, rather than drawing on the student's own knowledge and reasoning, are treated as indicators of higher offloading. Prompting patterns, referring to the types of content students include in their prompts and how these are distributed across assignments, serve as the primary observable trace of how cognitive effort is allocated during collaboration (Geng et al., 2026; López-Pernas et al., 2025a; Urban et al., 2025).

3. Methods

This section is organized around the three methodological steps used to collect and analyze the data. The data collection procedure is described first, followed by qualitative coding of the prompts and their validation, and then the Co-Occurrence Network Analysis is presented.

3.1. Data collection

The data used in this study were collected during a "Social Network Analysis" course at the University of Eastern Finland, an elective course for Bachelor's and Master's students in Information Technology. The course covers the fundamentals of network analysis, including network construction, visualization, statistical analysis, and interpretation. As part of the course, students completed four assignments in which they used an LLM of their choice to generate a network dataset that met the specified criteria. The instructions for each LLM-based assignment were as follows:

1. Use an LLM to create a sample network of your choice (e.g., food recipes, transport, interactions between countries, messages between people). The network should have at least 20 nodes and 50–100 edges. Please make sure that the nodes are connected and that there are some interactions.
2. Use an LLM to generate a network with 30–50 nodes. Ask it to make a certain node central. Use variations and follow up to get better results.
3. Choose one network from the course literature reading materials and try to generate it using an LLM.
4. Using an LLM, describe the network chosen for your group project within the prompt and try to get the LLM to resemble it. The network should have at least 30 nodes.

Attaching the conversations with LLMs was required for students' assignment submissions, along with a short written document explaining the network analysis and the LLM-generated dataset.

Students were free to use any LLM of their choice. Each assignment was graded on a scale from 1 to 10, based on the suitability of the generated dataset and the student's proficiency in plotting networks. Grades were assignment-based rather than student-based, meaning a single student could receive different quality classifications across assignments. The grade distribution was strongly skewed toward the maximum, with a mean of 9.48, a median of 10, and an SD of 1.24, indicating that most submissions fully met the task criteria. Submissions receiving the maximum score were therefore classified as *high-quality*, while all others were classified as *low-quality*, reflecting a natural distinction between submissions that fully met the task criteria and those that did not.

A total of 61 students were enrolled in the course, with each student completing up to four assignments. Not all assignment submissions included usable LLM interaction data: some students did not attach their chat logs, while others submitted interactions in formats that could not be analyzed, such as low-resolution screenshots or incomplete exports. After excluding submissions with unusable chat logs, student–AI conversations from 122 submissions were retained for analysis, yielding 281 prompts that were extracted and coded. On average, each conversation contained 2.81 prompts ($SD = 2.83$, $Median = 2$), with only 48% of the conversations consisting of more than one prompt.

3.2. Qualitative data coding and validation

The data were coded using each individual prompt as the unit of analysis. Prompts were binary-coded for the presence/absence of a code, and a single prompt could be assigned multiple codes. To guide coding, a hybrid approach was employed that integrated a top-down coding scheme with adaptations informed by data insights. The codebook developed by Giray (2023), which describes different elements of a prompt, was adapted for this purpose (see Table 1). The coding scheme was validated by having two researchers independently code a sample of student prompts ($n = 105$). Cohen's Kappa reliability indicated at least substantial agreement for each code ($\kappa > 0.60$), after which the remaining prompts were coded by a single researcher.

3.3. Co-occurrence network analysis

Co-Occurrence Network Analysis (CNA) is a network-based method within the Transition Network Analysis (TNA) toolkit (Saqr, López-Pernas et al., 2025) that identifies how states or codes appear together within a defined window of utterances. Within this window, any states or codes that occur simultaneously are counted as co-occurring. The co-occurrence counts depend on the selected *actor* parameter, which specifies the unit of analysis, such as an utterance, chat, or lesson, within which connections are counted (Misiejuk et al., 2026). In a CNA graph, nodes represent the prompt codes, and the edge weights indicate the cumulative frequency with which those codes co-occur within the same unit of analysis: thicker edges indicate more frequent co-occurrence, and the absence of an edge indicates that two codes never appeared together within the same unit. CNA treats connections as non-directional, capturing whether two codes co-occur without considering the sequence in which they appear. This contrasts with Frequency-based Transition Network Analysis (FTNA) (Saqr, Misiejuk et al., 2025), which also counts connections but preserves their directional order.

CNA is particularly suited to the present study because cognitive offloading is operationalized through the co-occurrence of prompt codes rather than their isolated frequency. Unlike simple frequency

Table 1
Prompt coding scheme with code frequencies, definitions, and example prompts.

Category	n	Definition	Example prompts
Instruction	249	Prompts that directly address the AI and provide general instructions	"Can you paste it to me as a csv raw text?" "Convert that into a unipartite network..."
Context	115	Provides contextual information about the network	"A network that illustrates the artist collaboration within the Nordics". "Generate a network of 30 major historical events from the Warring States Period (475 BC - 221 BC)"
Output specification	239	Outlines specific network requirements	"Can you create me a food recipes network that should have at least 20 nodes and 70 edges?" "I need only 30 different students. Make Student 10 have the highest degree centrality".
Disagreement	57	Explicitly disagrees with AI's output or asks the AI to correct or revise the output	"You are saying again and again to me to add while I am asking you to add the rest of the columns" "What about student-to-student interactions, it seems there is no kind of interactions in your provided edge lists?"
Agreement	15	Explicitly agrees with or acknowledges the AI's output	"Very good, make the csv file as an Excel table, please" "Good, now you elongate it to like 40 nodes"
Polite language	58	Explicitly uses polite or positive language	"Kindly use arbitrary names". "Please try to generate an edgelist based on the context"

counts, which collapse prompting behavior into aggregate totals, CNA preserves the relational structure of code combinations, mapping how prompt codes co-occur within individual student–AI interactions and accumulating these connections across all interactions within each comparison group or assignment. The strength of co-occurrence edges directly maps onto the construct of interest: stronger connections between *context* and *instruction* reflect higher cognitive engagement, as these codes require the student to draw on their own domain knowledge and task understanding, whereas *disagreement* and *agreement*, being reactive codes triggered by AI output rather than independent reasoning, reflect higher offloading. *Output specification* occupies an intermediate position, indicating task awareness and planning but delegating the actual generative work to the AI, and *polite language*, while not a direct indicator of cognitive effort, may signal a more deliberate and effortful approach to the interaction. By constructing separate CNA networks for each assignment and comparing edge strengths between high- and low-quality submissions, it is possible to trace how prompting patterns develop longitudinally and whether differences between groups emerge from the outset or widen as task demands accumulate across the course.

The CNA analysis was performed using the `TNA` R package (Tikka et al., 2025). To address research question 1, the examination began with the number of prompts per chat. Next, a Pearson residual analysis was conducted using mosaic plots to identify codes that were over- or underrepresented in high- and low-quality submissions. Co-occurrence patterns between groups were compared using a z-test for proportions. Separate CNA models were then constructed for each submission group based on the raw co-occurrence frequencies of the codes, with the *actor* parameter set to a single prompt. To facilitate plot comparison, min–max scaling was applied to edge weights to normalize values, so that the smallest becomes 0 and the largest 1, with all others scaled proportionally in between. A single subtraction plot was then generated to directly compare connection strengths between high- and low-quality submissions. Finally, centrality analyses examined the diffusion and in-strength centralities of each code for both high- and low-quality submissions. Diffusion centrality measures how effectively a code can spread through the network, while in-strength centrality reflects the total weight of connections directed toward a code. To address research question 2, four subtraction plots were created using scaled edge values to compare co-occurrences between high- and low-quality submissions for each assignment. Finally, the Pearson correlation of connection patterns between high- and low-quality submissions across assignments was compared.

4. Results

This section is structured around the two research questions posed in this study. The first research question examines which prompt-code combinations and centrality patterns characterize high- and low-quality submissions and what they reveal about cognitive offloading. The second question addresses whether prompt code co-occurrence patterns develop differently across four successive assignments and whether the two groups diverge over time.

Addressing the first research question, the analysis begins with prompt frequency before examining code co-occurrence structure within prompts. Nearly half of the chats in the dataset consisted of a single prompt per chat ($52\%, n = 64$). Among chats with multiple prompts, 40% contained two prompts ($n = 24$), 22% contained three prompts ($n = 13$), and 15% contained four prompts ($n = 9$). The remaining chats ranged from 5 to 10 prompts, with one containing as many as 18. The standardized residual analysis has shown only one statistically significant result: *disagreement* was significantly underrepresented in high-quality submissions and overrepresented in low-quality submissions (see Fig. 1). The sequence comparison analysis showed that the pattern *disagreement–instruction* occurs significantly more often in low-quality submissions (7.1%) than in high-quality submissions ($1.9\%; z = 5.93, p = 0.015$) (see Fig. 1(b)). Most other patterns, including *output specification–context*, *output specification–polite language*, and *instruction–output specification–context*, show differences in proportions between groups but are not statistically significant.

To examine how prompt codes co-occur within student–AI conversations, CNA was applied to reveal differences in prompting between high- and low-quality submissions (see Figs. 2(a) and 2(b)). In the CNA graphs, nodes represent prompt codes and edge thickness reflects the cumulative co-occurrence frequency between two codes within the same prompt. In the subtraction networks, red edges indicate stronger co-occurrences in high-quality submissions and blue edges indicate stronger co-occurrences in low-quality submissions, with edge thickness proportional to the difference in scaled co-occurrence values between the two groups. Nodes with no connecting edge did not co-occur within the same prompt for a specific group (high- or low-performing students). In submissions of both quality levels, the strongest connection is between *output specification* and *instruction* (low-quality n = 88; high-quality n = 122). The next most frequent relationships are also shared across both groups: *instruction–context* (low-quality n = 41; high-quality n = 71), followed by *output specification–context* (low-quality n = 39; high-quality n = 70). Differences become apparent beyond these core connections. In low-quality submissions, disagreement-related features are more prominent, with *output specification–disagreement* (n = 28) and *disagreement–instruction* (n = 21) standing out, while *instruction–polite language* occurs less frequently (n = 15). In high-quality submissions, these disagreement-based links are replaced by tone-related ones: *instruction–polite language* (n = 40), *output specification–polite language* (n = 35), and *context–polite language* (n = 35). The pattern of missing connections also differs. Low-quality submissions show no co-occurrence between *polite language–agreement*, *disagreement–context*, and *disagreement–agreement*. In contrast, high-quality submissions lack only *context–agreement* and *disagreement–context*.

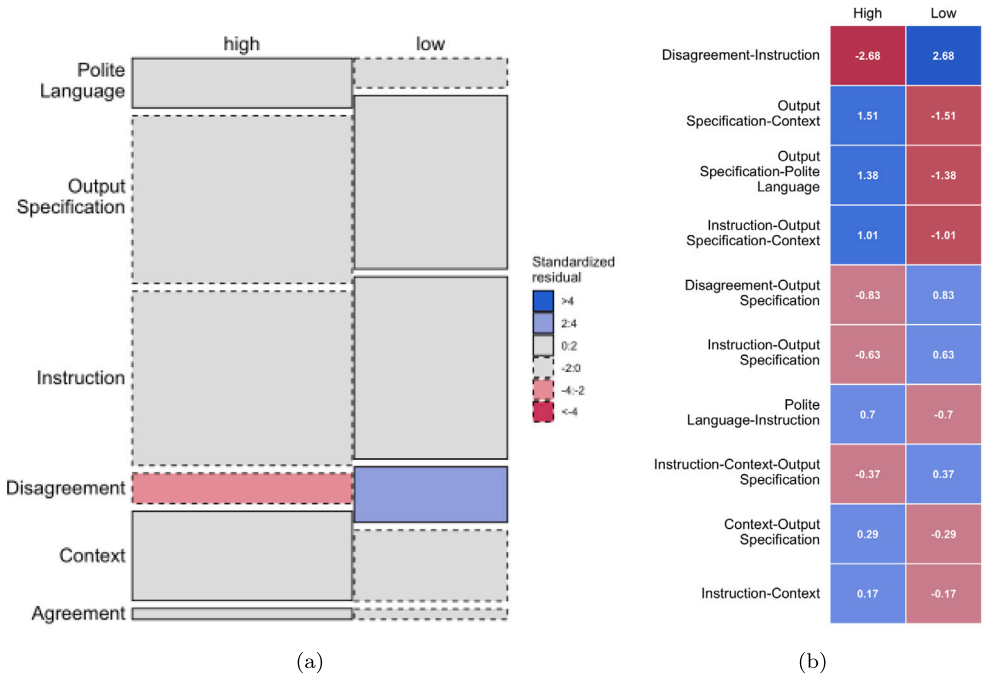


Fig. 1. (a) Mosaic plot showing the association between prompt features and submission quality. Standardized residuals highlight statistically significant patterns: overrepresented features are shown in blue, underrepresented features in red, and gray indicates no significant difference; (b) Co-occurrence comparison between high- and low-quality submissions (co-occurrences with a minimum of 5 frequencies). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

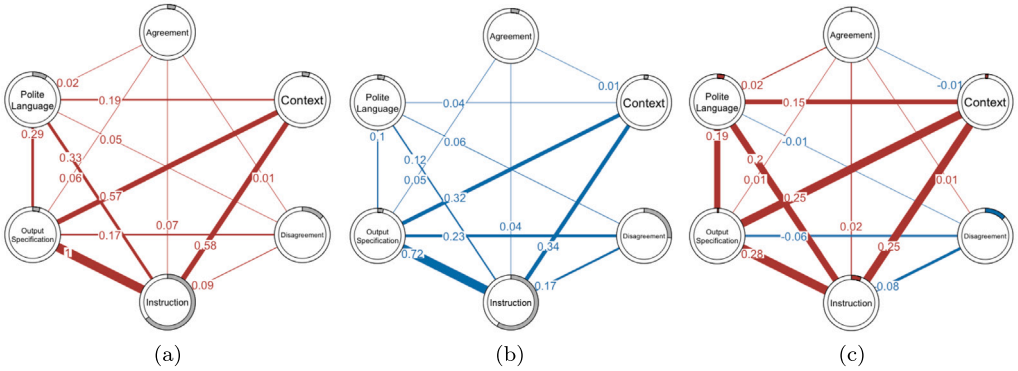


Fig. 2. (a) CNA model for high-quality submissions; (b) CNA model for low-quality submissions; (c) CNA subtraction model (stronger connections for high-quality submissions in red, for low-quality submissions in blue. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Complementing the individual networks, the CNA subtraction network quantifies which prompt code combinations most strongly differentiate high- and low-quality submissions in terms of cognitive offloading (see Fig. 2(c)). In low-quality submissions, stronger connections appear among disagreement-related and core structural features: *output specification–disagreement* (0.06), *instruction–disagreement* (0.08), and *polite language–disagreement* (0.01). Additionally, a slightly stronger connection was observed between *agreement* and *context*, as this link was entirely absent in prompting in high-quality submissions. In high-quality submissions, stronger connections are observed among most codes in the network, and the differences are generally more pronounced than the dominant connections seen in low-quality submissions. In particular, stronger co-occurrence connections are observed among both structural and stylistic codes. *Polite language* frequently co-occurs with *context* (0.15), *instruction* (0.20), and *output specification* (0.19). Core structural elements also show strong co-occurrences, including *context–output specification* (0.25), *context–instruction* (0.25), and *output specification–instruction* (0.28).

The centrality analysis completes the picture for the first research question by examining how prominently each code features in the overall co-occurrence structure of a prompt through in-strength and diffusion metrics. High-quality submissions exhibit substantially higher in-strength across all prompt elements, particularly *instruction* and *output specification* (see Fig. 3). In contrast, low-quality submissions show much lower in-strength centrality, while the relative importance of elements like *context*, *agreement*, and *polite language* remains similar. Furthermore, high-quality submissions show substantially higher diffusion centralities for key prompt elements, especially for *instruction* and *output specification* (see Fig. 3). In contrast, low-quality submissions exhibit much lower diffusion across all elements.

The second research question examines how prompt code co-occurrence patterns developed across the four assignments and whether patterns associated with high- and low-quality submissions diverged over time. Separate CNA networks were constructed for each assignment for both submission quality groups, and subtraction networks were generated to compare co-occurrence strengths between codes for

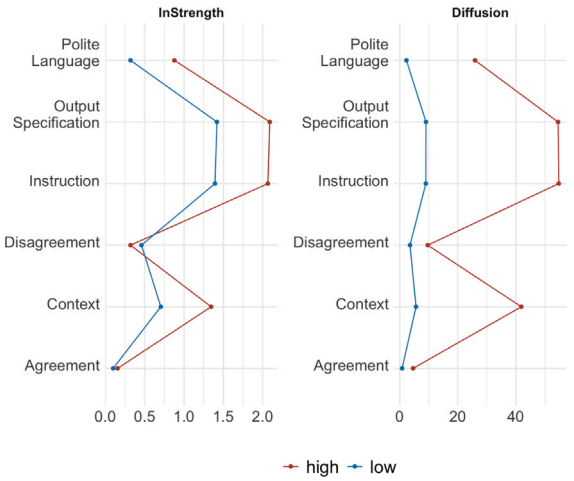


Fig. 3. Centrality analysis for diffusion and in-strength across codes.

high-and low-quality submissions. Across low-quality assignments, *output specification–instruction* consistently show the strongest connections, often accompanied by links to *context* (*output specification–context*; *instruction–context* (see Fig. 4). *Disagreement*-related features become more prominent in later assignments, forming additional connections with *output specification* and *instruction*. *Agreement* connections are generally sparse, emerging gradually across assignments with low counts. In contrast, connections from *polite language* decrease over time in both frequency and variability, initially targeting multiple features—*instruction* (n = 7), *output specification* (n = 7), *disagreement* (n = 5), and *context* (n = 3)—but later assignments show fewer and weaker links.

Analysis of high-quality submissions shows that the connection between *output specification* and *instruction* remains the strongest in every assignment, supported by consistently strong *output specification–context* and *instruction–context* connections. *Polite language*-related connections, particularly *output specification–polite language*, are consistently present but decline over time from n = 12–14 in early assignments to n = 2 in Assignment 4. *Disagreement*-related features show more engagement compared with low-quality submissions, with multiple outgoing connections in assignments 2 and 3, targeting *output specification*, *instruction*, *polite language*, and occasionally *agreement*, but these connections are weaker or less frequent in Assignments 1 and 4.

These patterns are also reflected in the CNA comparison networks. Across the four assignments, the CNA networks show a clear and widening divide in prompting patterns between high- and low-quality submissions. In Assignments 1 through 3, the co-occurrences are consistently stronger for high-quality submissions, with particularly strong connections among *instruction*, *output specification*, *context*, and *polite language*. Certain edges remain consistently stronger for high-quality submissions across all assignments, such as *context–output specification*, although their magnitude decreases slightly over time. Similarly, *polite language* maintains stronger co-occurrences for high-quality submissions with other key prompt features throughout the assignments. In contrast, low-quality submissions are primarily characterized by connections involving *disagreement*. This trend is most pronounced in the final assignment, where the *disagreement–instruction* and *disagreement–output specification* links are stronger for the low-quality group, with values of 0.391 and 0.445. No edges are stronger for low-quality submissions in Assignments 2 and 3.

The correlation values between the two groups' co-occurrence patterns across assignments provide a quantitative summary of whether prompting strategies converged or diverged over time. The correlation values reinforce the developmental pattern found in the CNA analysis. Assignments 1 through 3 show high similarity between the connection patterns of high- and low-quality submissions, with Pearson correlations of 0.952, 0.935, and 0.982 and Spearman correlations of 0.770, 0.890, and 0.912. In Assignment 4, the similarity drops, with Pearson's $\rho = 0.485$ and Spearman's $\rho = 0.485$.

5. Discussions

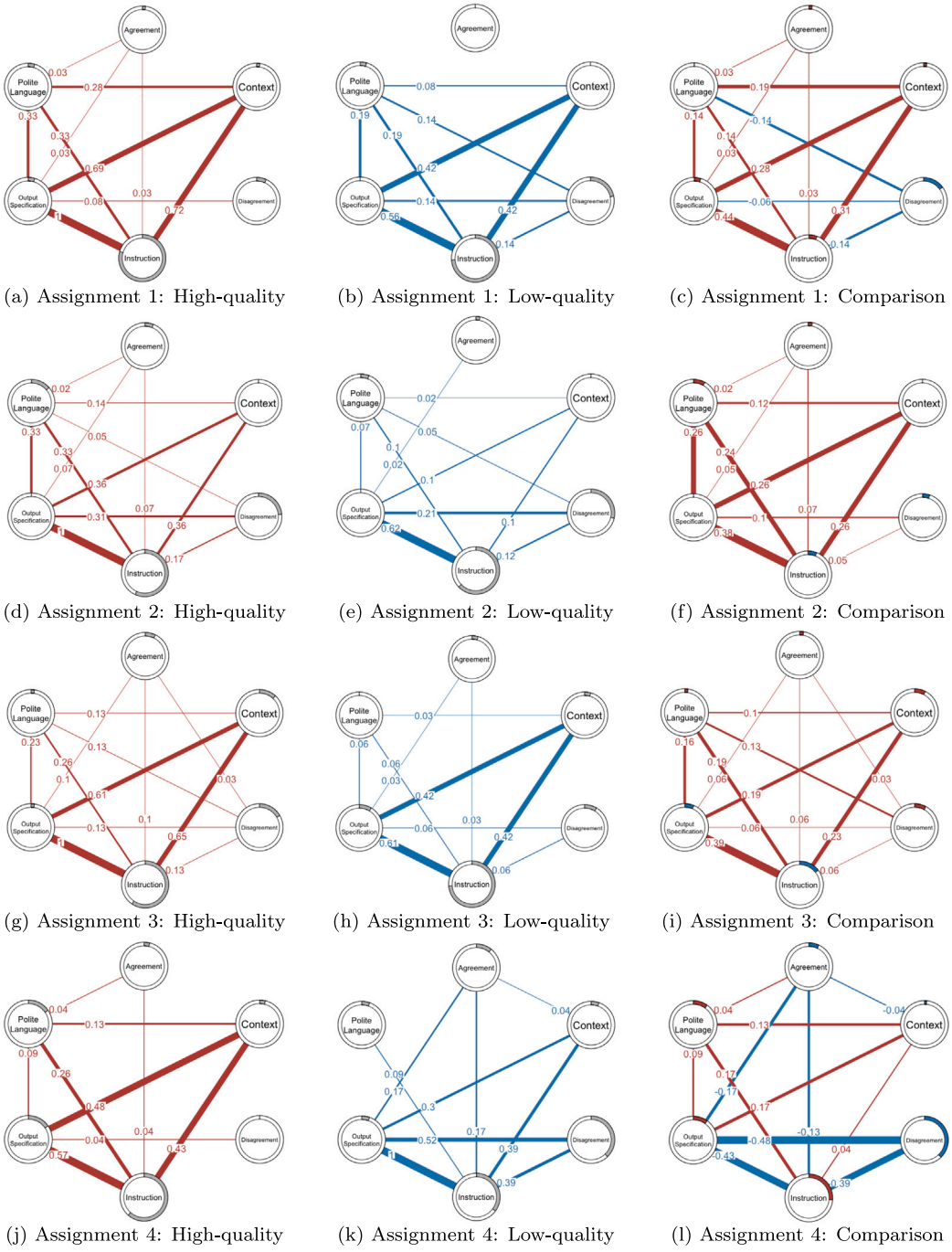
The emergence of LLMs has created new opportunities for completing assignments, but questions remain about how these tools can be meaningfully integrated into teaching and learning. Using LLMs requires students to develop new skills, such as crafting effective prompts, and involves decisions about how much cognitive effort to allocate to the task versus offloading to the AI (Misiejuk et al., 2025). In the present study, student–AI interactions during four assignments focused on generating social network datasets were examined. The findings highlight how students navigated this balance of cognitive offloading and engagement.

5.1. Prompting patterns and cognitive offloading in high- and low-quality submissions

This study reveals systematic differences in how students structured their prompts. High-quality submissions increasingly integrate polite, constructive language with instructional and contextual features, forming cohesive, balanced networks over time. Students include contextual details and specifications and do not delegate the interpretation of their intent to the LLM, which resulted in fewer instances of disagreement and more cohesive prompts. In contrast, low-quality submissions were characterized by disagreement and direct instructions, with fewer details or signs of satisfaction, e.g., polite language (thank-yous or polite requests). Direct instructional connections remain strong in both groups, but the networks in high-quality submissions demonstrate greater cohesion and integration, whereas those in low-quality submissions are more polarized and concentrated.

Notwithstanding these differences between high- and low-quality submissions, the overarching picture is dominated by task-oriented instructions that placed greater emphasis on producing the outcome than on developing thoughtful, well-structured prompts. This likely reflects how LLMs work: students expect high-quality outputs from minimal input, leading them to overlook the importance of domain knowledge, context, and reasoning in prompt construction. This pattern can be partly attributed to the way LLMs invite cognitive offloading: students quickly learn that models may infer missing context or correct under-specified tasks. As a result, prompting becomes an exercise in practicing a low-cognitive task, where minimal effort is sufficient to produce an acceptable output. This dynamic may inadvertently foster a form of cognitive "laziness", in which students rely on the model's generative capabilities to understand their intent rather than engage in more thoughtful processes of reasoning, planning, and articulating precise specifications (Dawson, 2020). Ultimately, the ease afforded by LLMs may come at the cost of the reasoning capacities that authentic learning is meant to cultivate. From the perspective of the learning sciences, this pattern aligns with longstanding concerns about how tools that reduce cognitive effort can unintentionally undermine deeper forms of learning. Research on cognitive load theory highlights that while reducing extraneous effort can be beneficial, excessive offloading risks diminishing the germane cognitive load necessary for developing transferable mental models and domain expertise (Carter, 2018, 2020).

A leveling tendency emerged when examining students' effort and iterative behaviors. Across both groups, a convergence toward low-effort, direct instruction was observed, with only a few submissions demonstrating deeply considered prompts. AI appeared to equalize achievement and effort levels, fostering an approach in which most students, regardless of ability, accepted the output they received with minimal modification or critical engagement. The convenience of generative AI encouraged a form of cognitive laziness among high achievers while reinforcing low performers' inertia (López-Pernas & Saqr,



(a) Assignment 1: High-quality (b) Assignment 1: Low-quality (c) Assignment 1: Comparison

(d) Assignment 2: High-quality (e) Assignment 2: Low-quality (f) Assignment 2: Comparison

(g) Assignment 3: High-quality (h) Assignment 3: Low-quality (i) Assignment 3: Comparison

(j) Assignment 4: High-quality (k) Assignment 4: Low-quality (l) Assignment 4: Comparison

Fig. 4. CNA models by assignment and submission quality; stronger connections in high-quality submissions are red, in low-quality submissions are blue. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

2024). Because AI produced plausible answers instantly and was adept at understanding students' intent, it reduced the incentive for refinement and deeper cognitive effort, effectively lowering the standard of engagement to the cohort's minimal common denominator. Whether this leveling tendency reflects a general characteristic of LLM-assisted task completion or is specific to the task design and student population examined here remains an open question. Future work could examine whether structured prompting scaffolds or tasks that require domain knowledge the AI cannot infer independently would reduce this convergence toward minimal engagement. The dominance of instruction-only prompts across both groups aligns with research on the effects of using LLMs on cognitive and metacognitive processes, which is still emerging. For example, Stadler et al. (2024) found evidence that using LLMs reduces mental effort, which in turn can lead to lower-quality reasoning and argumentation. Similarly, Sabqat et al. (2025) reported that medical students showed high reliance on AI and low motivation for independent analysis, based on survey data. In contrast, Fan et al. (2025) found no significant performance differences between the AI-supported group and groups with no tools or human support; however, the AI-supported group exhibited fewer metacognitive processes, such as evaluation and orientation. A similar pattern of reduced engagement in surface-level regulation and limited activation of deeper learning processes was also observed in programming education (López-Pernas et al., 2025a).

5.2. Longitudinal development of prompting patterns

There are also important differences in how interaction strategies evolved over time. Across the four assignments, the prompting patterns of high- and low-performing students diverged rather than converged. High performers gradually refined their strategies, developing cohesive prompt structures in which contextualization and instruction were tightly integrated, an approach that remained robust even as superficial elements, such as polite markers, diminished. Over the course of the assignments, polite language becomes more prominent in high-quality submissions, indicating a positive attitude resulting from appreciation for the results. Low performers, however, did not move toward this effective model; instead, their prompts displayed increasing levels of disagreement and reactive correction, particularly by Assignment 4. In contrast, low-quality submissions had more disagreement instances, which became more central and persistent.

This divergence suggests that reliance on LLMs and associated cognitive offloading does not diminish with repeated use. Rather, it may reflect a tendency that, over the course of a single semester, shows little sign of self-correction without intervention. It should be noted that this interpretation is tentative: the observed divergence may also reflect increasing task demands across assignments rather than the consolidation of stable prompting dispositions (Sultanova, 2025). Familiarity with the tool may additionally reduce novelty-driven effort over time, contributing to the convergence toward minimal prompting observed across both groups in later assignments (Wang & Luo, 2026).

The results of this study align with those of Zhang et al. (2025), who reported that AI often mirrored students' existing states and dispositions rather than improving them. In their study, high-performing students collaborated more effectively with AI, using it for strategic planning and optimization, whereas low-performing students showed greater resistance and relied more heavily on manual processes. The sharpest divergence in Assignment 4, where low-quality submissions became increasingly disagreement-centered, is particularly consistent with this pattern. Furthermore, the study by Shen et al. (2025) showed that even when students were taught a four-step strategy, they quickly shifted from resolving recognition conflicts to using the GenAI system primarily for content generation. Without explicit scaffolding or targeted AI literacy training, struggling students appear to revert to inefficient trial-and-error strategies, thereby reinforcing rather than alleviating their difficulties. Early evidence suggests that scaffolding is essential.

6. Implications

Student–AI collaboration to solve tasks reshapes how learners plan, monitor, and evaluate their work, highlighting the interplay between cognitive regulation, co-regulation, and cognitive load. Cognitive regulation involves deliberately managing thinking, motivation, emotions, and behavior to achieve learning goals (Järvelä & Hadwin, 2013) and becomes especially important in AI-mediated contexts, where learners must actively engage with LLMs rather than rely on them passively. Prompting an LLM requires planning, providing relevant input, evaluating outputs, and refining prompts, with motivational and emotional regulation supporting sustained effort.

Co-regulation occurs through interactions with peers, teachers, and AI tools, which offer feedback, elaboration, or alternative solutions that learners can integrate into their regulatory processes (Järvelä & Hadwin, 2013; J.M. Lodge et al., 2023). Cognitive load interacts closely with these processes: high load can trigger monitoring and control, while regulation itself requires mental effort unless automatized (Klar, 2025). In AI-supported learning, LLMs can reduce load by offloading routine tasks or increase it through active prompting and evaluation, illustrating dynamic cognitive offloading. Designing tasks that balance offloading with engagement is, therefore, critical to promoting deep learning and long-term retention.

The quality of student prompts reflects their regulatory engagement. High-quality prompts show forethought, contextual detail, and clear specifications, while low-quality prompts rely on the AI to fill in gaps, reflecting weaker planning and monitoring (Zimmerman & Schunk, 2011). Frequent disagreements in low-quality prompts indicate monitoring without effective revision, whereas high-performing students iteratively refine prompts, demonstrating productive evaluative judgment (Bearman et al., 2024; Tai et al., 2018).

The finding that disagreement was overrepresented in low-quality submissions and became increasingly central across assignments suggests that early intervention in prompting behavior may be more effective than corrective feedback after patterns have consolidated. Emerging empirical evidence shows that prompt quality can predict the quality of LLM outputs (Knoth et al., 2024), which indicates that integrating explicit guidance on prompt construction, and particularly on the value of supplying contextual framing and output specifications, into task introductions rather than treating these as supplementary AI literacy skills may directly improve the outcomes of students' AI-assisted work.

The longitudinal divergence between groups, with the sharpest difference emerging in Assignment 4, further suggests that a single snapshot of prompting behavior would not have been sufficient to capture how students' engagement with AI develops over time. Longitudinal monitoring of prompting patterns across a course could allow instructors to identify students whose strategies are becoming increasingly reactive before these tendencies consolidate.

Overall, learning with AI does not automatically result in effective cognitive offloading. Learners may offload too much, too little, or experience overload while interacting with the AI. These findings highlight the importance of designing tasks that balance the distribution of cognitive effort between the student and the AI. Providing opportunities for students to engage actively while supporting the offloading of routine or repetitive work can help maintain optimal mental effort, enhance engagement, and promote meaningful learning outcomes (Klar, 2025). Effective student–AI collaboration, therefore, requires attention to both the regulation of cognitive effort and the design of prompts and tasks that foster reflection, monitoring, and iterative refinement.

7. Limitations

This study is limited by its context: a single course on Social Network Analysis focusing on synthetic data generation. The specific nature of the task (generating structured datasets) may prioritize certain prompting elements (e.g., strict output specifications) more than other task types would. As such, future studies should replicate this investigation across diverse disciplines, task types (e.g., creative writing, open-ended problem solving), and learning environments to determine whether the observed patterns of cognitive offloading generalize beyond structured data generation tasks. Furthermore, while this study provides an initial exploration, its robustness could be strengthened by larger, more diverse samples, longitudinal designs to track cognitive offloading over time, and the inclusion of control groups to isolate the specific effects of LLM-based prompting on learning outcomes. Additionally, student prompts were used as a proxy for cognitive offloading; however, prompts alone may not fully capture the cognitive processes underlying student–AI interaction, and future studies should consider complementary measures such as think-aloud protocols or self-reported cognitive load. Furthermore, variability in LLM choice was not controlled for, as different students may have used different models, which could have influenced the nature and quality of the outputs received. The definition and measurement of submission quality were based solely on course grades, which, while practical, may not fully reflect the complexity of student learning or cognitive engagement. Finally, the co-occurrence networks employed in this study are non-directional, limiting insight into the sequential nature of prompting behavior; future studies should consider sequential or temporal analysis

methods to better capture the dynamic unfolding of student–AI collaboration. Finally, although disagreement was analyzed, the sentiment of the disagreement (e.g., frustration vs. constructive critique) was not qualitatively examined. Future studies should incorporate sentiment analysis of student–AI disagreements to better understand the emotional and cognitive dimensions of such interactions and their potential impact on learning outcomes.

8. Conclusions

This study examined cognitive offloading in student–AI collaboration through prompt co-occurrence patterns across four assignments, finding that high-quality submissions were characterized by prompts that combined contextual framing, task specifications, and polite language, while low-quality submissions were dominated by disagreement-centered, reactive patterns that widened longitudinally. As AI becomes more deeply embedded in higher education, how students interact with AI systems matters more than it might seem; every prompt is a trace of their thinking, planning, and approach to learning. These findings demonstrate that prompt co-occurrence patterns provide a traceable, longitudinal indicator of cognitive engagement, and future work should examine whether early prompting patterns predict later performance and whether scaffolding interventions can shift reactive tendencies before they consolidate.

CRediT authorship contribution statement

Kamila Misiejuk: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Sonsoles López-Pernas:** Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization. **Rogers Kaliisa:** Writing – review & editing, Writing – original draft, Data curation, Conceptualization. **Mohammed Saqr:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

References

An, S., Zhang, S., Guo, T., Lu, S., Zhang, W., & Cai, Z. (2025). Impacts of generative AI on student teachers' task performance and collaborative knowledge construction process in mind mapping-based collaborative environment. *Computers & Education*, *227*, Article 105227. http://dx.doi.org/10.1016/j.compedu.2024.105227.

Awalurahman, H. W., Aji, R. F., & Budi, I. (2025). Transformer and large language models for automatic multiple-choice question generation: A systematic literature review. *IEEE Access*, *13*, http://dx.doi.org/10.1109/ACCESS.2025.3590423.

Baidoo-Anu, D., & Owusu Ansah, L. (2023). Education in the era of generative artificial intelligence (AI): Understanding the potential benefits of ChatGPT in promoting teaching and learning. *Journal of Aircraft*, *7*, 52–62. http://dx.doi.org/10.61969/jai.1337500.

Bearman, M., Tai, J., Dawson, P., Boud, D., & Ajjawi, R. (2024). Developing evaluative judgement for a time of generative artificial intelligence. *Assessment & Evaluation in Higher Education*, *49*(6), 893–905. http://dx.doi.org/10.1080/02602938.2024.2335321.

Carter, J. A. (2018). Autonomy, cognitive offloading, and education. *Educational Theory*, *68*(6), 657–673. http://dx.doi.org/10.1111/edth.12338.

Carter, J. A. (2020). Intellectual autonomy, epistemic dependence and cognitive enhancement. *Synthese*, *197*(7), 2937–2961. http://dx.doi.org/10.1007/s11229-017-1549-y.

Chen, A., Xiang, M., Zhou, J., Jia, J., Shang, J., Li, X., Gašević, D., & Fan, Y. (2025). Unpacking help-seeking process through multimodal learning analytics: A comparative study of ChatGPT vs Human expert. *Computers & Education*, *226*, Article 105198. http://dx.doi.org/10.1016/j.compedu.2024.105198.

Cong-Lem, N., & Thuy-Dung, N. T. (2026). Towards a framework for understanding and assessing critical thinking in generative AI-enhanced learning environments: A scoping review. *European Journal of Psychology of Education*, *41*(2), 41. http://dx.doi.org/10.1007/s10212-026-01089-y.

Dawson, P. (2020). Cognitive offloading and assessment. In *Re-imagining university assessment in a digital world* (pp. 37–48). http://dx.doi.org/10.1007/978-3-030-41956-1_4.

Elsayed, H. (2024). The impact of hallucinated information in large language models on student learning outcomes: A critical examination of misinformation risks in AI-assisted education. *Northern Reviews on Algorithmic Research, Theoretical Computation, and Complexity*, *9*(8), 11–23.

Fan, Y., Tang, L., Le, H., Shen, K., Tan, S., Zhao, Y., Shen, Y., Li, X., & Gašević, D. (2025). Beware of metacognitive laziness: Effects of generative artificial intelligence on learning motivation, processes, and performance. *British Journal of Educational Technology*, *56*(2), 489–530. http://dx.doi.org/10.1111/bjet.13544.

Geng, F., Shah, A., Li, H., Mulla, N., Swanson, S., Soosai Raj, G., Zingaro, D., & Porter, L. (2026). Exploring student-AI interactions in vibe coding. In *Proceedings of the 28th Australasian Computing Education Conference* (pp. 45–54). http://dx.doi.org/10.1145/3786228.3786236.

Giray, L. (2023). Prompt engineering with ChatGPT: A guide for academic writers. *Annals of Biomedical Engineering*, *51*, 2629–2633. http://dx.doi.org/10.1007/s10439-023-03272-4.

Grinschgl, S., Papenmeier, F., & Meyerhoff, H. S. (2021). Consequences of cognitive offloading: Boosting performance but diminishing memory. *Quarterly Journal of Experimental Psychology*, *74*(9), 1477–1496. http://dx.doi.org/10.1177/17470218211008060.

Gu, C., Zhang, W., Huang, Z., Kou, J., Liu, Z., Zhao, C., Liu, C., Zhang, L., Lin, W., Wang, Z., et al. (2024). LENS: Layers of evaluation of hallucination in genai systems. In *2024 7th International Conference on Universal Village* (pp. 1–85). IEEE, http://dx.doi.org/10.1109/UV63228.2024.11189150.

Hao, Z., Cao, J., Li, R., Yu, J., Liu, Z., & Zhang, Y. (2026). Mapping student-AI interaction dynamics in multi-agent learning environments: Supporting personalised learning and reducing performance gaps. *Computers & Education*, *241*, Article 105472. http://dx.doi.org/10.1016/j.compedu.2025.105472.

Järvelä, S., & Hadwin, A. F. (2013). New frontiers: Regulating learning in CSCL. *Educational Psychologist*, *48*(1), 25–39. http://dx.doi.org/10.1080/00461520.2012.748006.

Joo, S., Han, I., & Park, I. (2026). The effect of metacognitive prompts using generative AI on cognitive load, task performance, and self-efficacy in online self-regulated learning. *Educational Psychology*, 1–24. http://dx.doi.org/10.1080/01443410.2026.2618717.

Kaliisa, R., Misiejuk, K., López-Pernas, S., & Saqr, M. (2026). How does artificial intelligence compare to human feedback? A meta-analysis of performance, feedback perception, and learning dispositions. *46*, (pp. 80–111). http://dx.doi.org/10.1080/01443410.2025.2553639.

Kim, N., & Ji, Y. G. (2026). Exploratory search with generative AI: An empirical study on the impact of interaction design strategies on information exploration and cognitive load. *International Journal of Human-Computer Studies*, Article 103771. http://dx.doi.org/10.1016/j.ijhcs.2026.103771.

Klar, M. (2025). *Generative AI chatbots for self-regulated learning while balancing cognitive load: Perceptions, interaction patterns, and instructional designs in k-12 learning* Dissertation, Universität Duisburg-Essen.

Knoth, N., Tolzin, A., Janson, A., & Leimeister, J. M. (2024). AI literacy and its implications for prompt engineering strategies. *Computers and Education: Artificial Intelligence*, *6*, Article 100225. http://dx.doi.org/10.1016/j.caeai.2024.100225.

Lai, J. W., Qiu, W., Thway, M., Zhang, L., Jamil, N. B., Su, C. L., Ng, S. S., & Lim, F. S. (2025). Leveraging process-action epistemic network analysis to illuminate student self-regulated learning with a socratic chatbot. *Journal of Learning Analytics*, *12*(1), 32–49. http://dx.doi.org/10.18608/jla.2025.8549.

Lane, R. (2025). Mitigating generative AI hallucinations in geographical education. *International Research in Geographical and Environmental Education*, 1–17. http://dx.doi.org/10.1080/10382046.2025.2555185.

Liu, J., Li, S., & Dong, Q. (2024). Collaboration with generative artificial intelligence: An exploratory study based on learning analytics. *Journal of Educational Computing Research*, *62*(5), 1014–1046. http://dx.doi.org/10.1177/07356331241242441.

Lodge, J. M., de Barba, P., & Broadbent, J. (2023). Learning with generative artificial intelligence within a network of co-regulation. *Journal of University Teaching and Learning Practice*, *20*(7), 1–10. http://dx.doi.org/10.3316/informit.T2024110800007291367503485.

Lodge, J., Thompson, K., & Corrin, L. (2023). Mapping out a research agenda for generative artificial intelligence in tertiary education. *Australasian Journal of Educational Technology*, *39*, 1–8. http://dx.doi.org/10.14742/ajet.8695.

López-Pernas, S., Misiejuk, K., Oliveira, E., & Saqr, M. (2025a). The dynamics of the self-regulation process in student-AI interactions: The case of problem-solving in programming education. In *Proceedings of the 25th koli calling international conference on computing education research* (pp. 1–12). http://dx.doi.org/10.1145/3769994.3770043.

López-Pernas, S., & Saqr, M. (2024). How the dynamics of engagement explain the momentum of achievement and the inertia of disengagement: A complex systems theory approach. *Computers in Human Behavior*, *153*, Article 108126. http://dx.doi.org/10.1016/j.chb.2023.108126.

Ma, N., & Zhong, Z. (2025). A meta-analysis of the impact of generative artificial intelligence on learning outcomes. *Journal of Computer Assisted Learning*, *41*(5), Article e70117. http://dx.doi.org/10.1111/jcal.70117.

Michel-Villarreal, R., Vilalta-Perdomo, E., Salinas-Navarro, D., Thierry-Aguilera, R., & Gerardou, F. (2023). Challenges and opportunities of generative AI for higher education as explained by ChatGPT. *Education Sciences*, *13*, 856.

Misiejuk, K., López-Pernas, S., Kaliisa, R., & Saqr, M. (2025). Mapping the landscape of generative artificial intelligence in learning analytics: A systematic literature review. *Journal of Learning Analytics*, *12*(1), 12–31. http://dx.doi.org/10.18608/jla.2025.8591.

Misiejuk, K., Oliveira, E. A., Song, Y., López-Pernas, S., & Saqr, M. (2026). Modeling observation lens in co-occurrence network analysis: Four window approaches and implementations. In *Proceedings of the Transition Network Analysis Workshop, April 27-28, 2026, 16th International Conference on Learning Analytics & Knowledge (LAK) Conference, Bergen, Norway*. http://dx.doi.org/10.2139/ssrn.6189298.

Molenaar, I., Horvers, A., & Baker, R. (2019). What can moment-by-moment learning curves tell about students' self-regulated learning? *Learning and Instruction*.

Nguyen, H., & Nguyen, A. (2025). Reflective practices and self-regulated learning in designing with generative artificial intelligence: An ordered network analysis. *Journal of Science Education and Technology*, *34*(5), 1178–1192. http://dx.doi.org/10.1007/s10956-024-10175-z.

Roschelle, J., & Singleton, C. (2008). Graphing calculators: Enhancing math learning for all students. In *International Handbook of Information Technology in Primary and Secondary Education. Springer International Handbook of Information Technology in Primary and Secondary Education, vol 20* (pp. 951–959). Springer, http://dx.doi.org/10.1007/978-0-387-73315-9_60.

Sabqat, M., Ain, N., Iqbal, S., & Khan, R. A. (2025). Shortcut to knowledge or shortcut to thinking? Investigating AI-induced metacognitive laziness in future doctors. *Annals of King Edward Medical University*, *31*, 146–154. http://dx.doi.org/10.21649/akemu.v31iSpl2.6096.

Sandoval-Medina, C., Arévalo-Mercado, C. A., Muñoz-Andrade, E. L., & Muñoz-Arteaga, J. (2024). Self-explanation effect of cognitive load theory in teaching basic programming. *Journal of Information Systems Education*, *35*(3), 303–312. http://dx.doi.org/10.62273/GMIV1698.

Saqr, M., López-Pernas, S., Törmänen, T., Kaliisa, R., Misiejuk, K., & Tikka, S. (2025). Transition network analysis: A novel framework for modeling, visualizing, and identifying the temporal patterns of learners and learning processes. In *Proceedings of the 15th International Learning Analytics and Knowledge Conference* (pp. 351–361). http://dx.doi.org/10.1145/3706468.3706513.

Saqr, M., Misiejuk, K., Törmänen, T., Kaliisa, R., Tikka, S., & López-Pernas, S. (2025). Frequency transition network analysis (FTNA). In *Proceedings of the 18th International Conference on Computer-Supported Collaborative Learning - CSCL 2025*. http://dx.doi.org/10.22318/cscl2025.901997.

Schulz, T., Knierim, M. T., & Weinhardt, C. (2025). How generative-AI-assistance impacts cognitive load during knowledge work: A study proposal. In *Information Systems and Neuroscience. NeuroIS 2024. Lecture Notes in Information Systems and Organisation, vol 66* (pp. 357–365). http://dx.doi.org/10.1007/978-3-031-71385-9_31.

Sharples, M. (2023). Towards social generative AI for education: Theory, practices and ethics. *Learning Disability Research & Practice*, *9*, 159–167.

Shen, W., Lin, X.-F., Liu, J., Liang, X., Chen, R., Lai, X., & Zheng, X. (2025). The process of undergraduates' collaboration with a generative artificial intelligence chatbot: Insights from conversation content and epistemic network analysis. *Journal of Computer Assisted Learning*, *41*(6), Article e70140. http://dx.doi.org/10.1111/jcal.70140.

Skulmowski, A. (2023). The cognitive architecture of digital externalization. *Educational Psychology Review*, *35*(4), 101. http://dx.doi.org/10.1007/s10648-023-09818-1.

Stadler, M., Bannert, M., & Sailer, M. (2024). Cognitive ease at a cost: LLMs reduce mental effort but compromise depth in student scientific inquiry. *Computers in Human Behavior*, *160*, Article 108386. http://dx.doi.org/10.1016/j.chb.2024.108386.

Sultanova, G. (2025). Assessment tasks in STEM education: integrating bloom's revised taxonomy and cognitive load theory. *Research in Science & Technological Education*, 1–19. http://dx.doi.org/10.1080/02635143.2025.2586564.

Sweller, J. (1994). Cognitive load theory, learning difficulty, and instructional design. *Learning and Instruction*, *4*(4), 295–312. http://dx.doi.org/10.1016/0959-4752(94)90003-5.

Sweller, J. (2011a). *Cognitive load theory Psychology of Learning and Motivation*, *55*, 37–76. http://dx.doi.org/10.1016/B978-0-12-387691-1.00002-8.

Tai, J., Ajjawi, R., Boud, D., Dawson, P., & Panadero, E. (2018). Developing evaluative judgement: enabling students to make decisions about the quality of work. *Higher Education*, *76*(3), 467–481. http://dx.doi.org/10.1007/s10734-017-0220-3.

Tikka, S., López-Pernas, S., & Saqr, M. (2025). Tna: An r package for transition network analysis. *Applied Psychological Measurement*, http://dx.doi.org/10.1177/01466216251348840.

Urban, M., Lukavskỳ, J., Brom, C., Hein, V., Svacha, F., Děchtěrenko, F., & Urban, K. (2025). Prompting for creative problem-solving: A process-mining study. *Learning and Instruction*, *99*, Article 102156. http://dx.doi.org/10.1016/j.learninstruc.2025.102156.

Wang, X., Liu, Q., Pang, H., Tan, S. C., Lei, J., Wallace, M. P., & Li, L. (2023). What matters in AI-supported learning: A study of human-AI interactions in language learning using cluster analysis and epistemic network analysis. *Computers & Education*, *194*, Article 104703. http://dx.doi.org/10.1016/j.compedu.2022.104703.

Wang, Y., & Luo, X. (2026). Generative AI as a cognitive co-participant: Disciplinary modulation of EFL academic reading load and motivation. *40*, In *Proceedings of the AAAI Conference on Artificial Intelligence* (pp. 40889–40897). http://dx.doi.org/10.1609/aaai.v40i48.42122.

Wang, F., Tang, X., & Yu, S. (2025). Cognitive outsourcing based on generative artificial intelligence: An analysis of interactive behavioral patterns and cognitive structural features. *Acta Psychologica Sinica*, *57*(6), 967–986. http://dx.doi.org/10.3724/sp.j.1041.2025.0967.

Woollaston, S., Flanagan, B., Ocheja, P., Toyokawa, Y., & Ogata, H. (2025). ARCHIE: Exploring language learner behaviors in LLM chatbot-supported active reading log data with epistemic network analysis. In *Proceedings of the 15th international learning analytics and knowledge conference* (pp. 642–654). http://dx.doi.org/10.1145/3706468.3706556.

Yavariabdi, A., Paudel, B., Carleton, T., & De Almeida, C. D. A. (2025). Generative AI in assessment and feedback generation in higher education: A systematic review. In *2025 International Conference on Education Technology and Computers* (pp. 361–371). IEEE, http://dx.doi.org/10.1109/ICETC66579.2025.11387416.

Zhang, F., Gou, J., Shen, K. N., Camarinha-Matos, L. M., & Wang, Z. (2025). Effects of AI teammates on learning behavior in human-AI collaboration environments: A perspective on self-regulated learning. *Education and Information Technologies*, *30*, 26801–26825. http://dx.doi.org/10.1007/s10639-025-13717-z.

Zhang, J., & Sun, D. (2025). A systematic review of generative artificial intelligence in education. In *2025 7th International Conference on Computer Science and Technologies in Education* (pp. 552–556). IEEE, http://dx.doi.org/10.1109/CSTE64638.2025.11092288.

Zhao, Y., Xu, Y., & Long, T. (2026). Understanding pre-service science teachers' collaborative discourse patterns in a genai integrated collective reflection: A network analytic approach. *Journal of Science Education and Technology*, *35*, 141–159. http://dx.doi.org/10.1007/s10956-025-10241-0.

Zhou, J., Zhang, J., Wan, R., Cui, X., Liu, Q., Guo, H., Shi, X., Fu, B., Meng, J., Yue, B., et al. (2025). Integrating AI into clinical education: Evaluating general practice trainees' proficiency in distinguishing AI-generated hallucinations and impacting factors. *BMC Medical Education*, *25*(1), 406. http://dx.doi.org/10.1186/s12909-025-06916-2.

Zimmerman, B. J., & Schunk, D. H. (2011). Self-regulated learning and performance: An introduction and an overview. In *Handbook of self-regulation of learning and performance* (pp. 15–26). Routledge, http://dx.doi.org/10.4324/9780203839010.ch1.